Part-Aware Multidimensional Consistency for Semisupervised Detection of Industrial Composite Objects in Remote Sensing Images

Qinxue He, Bo Cheng, Peng Huang, Xiaoping Zhang, and Chuan He

***Abstract*—Industrial composite facilities, such as thermal power plants (TPPs), iron and steel plants (ISPs), and cement plants (CPs), are major sources of global carbon emissions, making their accurate detection in high-resolution remote sensing imagery (RSI) critical for environmental monitoring and energy management. However, using traditional semisupervised object detection (SSOD) methods to detect these facilities remains challenging due to: 1) multipart and structurally heterogeneous layouts that make box-level supervision insufficient; 2) pseudolabel noise that can be amplified during semisupervised self-training; and 3) the tradeoff between pseudolabel quality and quantity across training stages. To address these problems, we propose part-aware multi-dimensional consistency framework (PMC), a semisupervised part-aware detection framework with multidimensional consistency constraints. PMC consists of three complementary components: a part-based K-means module (PKM) that automatically discovers discriminative subregions within pseudolabeled boxes for fine-grained part-aware representation learning without part annotations, a multidimensional double-check mechanism (MDCM) that regularizes features and predictions via multidimensional consistency to suppress unreliable pseudosupervision, and a dynamic top-K pseudo-label selection (DTK) strategy that adaptively balances pseudolabel quality and quantity during training. Extensive experiments on the Indus-CO dataset and the Indus-GF dataset, along with composite subsets derived from DIOR and DOTA, show that PMC consistently outperforms fully supervised and state-of-the-art semisupervised baselines under limited-label settings. Ablation studies further verify the complementary contributions of PKM, MDCM, and DTK, while qualitative results demonstrate robust localization across diverse target layouts.**

***Index Terms*—Industrial composite object detection, part-based models (PBMs), remote sensing imagery (RSI), semisupervised object detection (SSOD).**

I. Introduction

INDUSTRIAL composite facilities such as thermal power plants (TPPs), iron and steel plants (ISPs), and cement plants (CPs) play a pivotal role in global energy consumption and carbon emissions. Accurate detection and monitoring of these facilities in high-resolution remote sensing imagery (RSI) is therefore critical for environmental management, energy optimization, and carbon reduction policies [1], [2], [3], [4], [5]. While modern detectors have achieved strong performance on generic object categories through advanced feature representation and localization strategies, such as high-resolution feature enhancement for small-object detection and dynamic instance mining for weakly supervised detection [6], [7], industrial composite facilities remain challenging because they are multipart and structurally heterogeneous. Recent studies have also explored more accurate localization strategies for remote sensing objects, such as dense representative point-guided rotated detection for ships [8]. However, these methods mainly focus on fully supervised detection or specific object categories, and the problem of semisupervised detection for complex industrial composite objects remains insufficiently explored. For example, as shown in Fig. 1, a TPP typically contains visually distinctive substructures such as boiler buildings, turbine halls, cooling towers, and chimneys; an ISP often includes blast furnaces, sintering plants, steelmaking workshops, and large material yards; and a CP generally consists of rotary kilns, preheater towers, coolers, clinker reservoirs, cement silos, and preblending domes. Under occlusion, scale variation, and cluttered backgrounds, models trained with only whole-instance bounding boxes often localize the coarse extent but fail to capture internal discriminative cues, leading to missed detections and unstable predictions [9].

To address the structural complexity of composite facilities, part-based models (PBMs) represent objects as a combination of a root object and multiple discriminative parts [10]. PBMs can improve robustness to occlusion and deformation [11], [12], but many existing approaches rely on dense part annotations or predefined part layouts, which are expensive and may not generalize to industrial facilities with diverse and flexible substructures. Moreover, in typical RSI datasets, only whole-instance bounding boxes are available; such box-level supervision does not explicitly encourage learning fine-grained internal structures, which limits the benefit of part modeling in practice.

Meanwhile, semisupervised object detection (SSOD) has emerged as a promising solution to reduce labeling costs by leveraging abundant unlabeled data via pseudolabeling and consistency regularization [13], [14], [15], [16], [17], [18], [19]. However, for industrial composite facilities, pseudolabel errors are more likely to occur (e.g., partial coverage,

Received 17 March 2026; revised 21 July 2026; accepted 15 September 2026. Date of publication 21 September 2026; date of current version 12 October 2026. This work was supported by Hainan Province Science and Technology Special Fund, the aerospace special fund for Hainan Aerospace Technology Innovation Center under Grant ATIC-202501003. *(Corresponding authors: Bo Cheng; Xiaoping Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3735722

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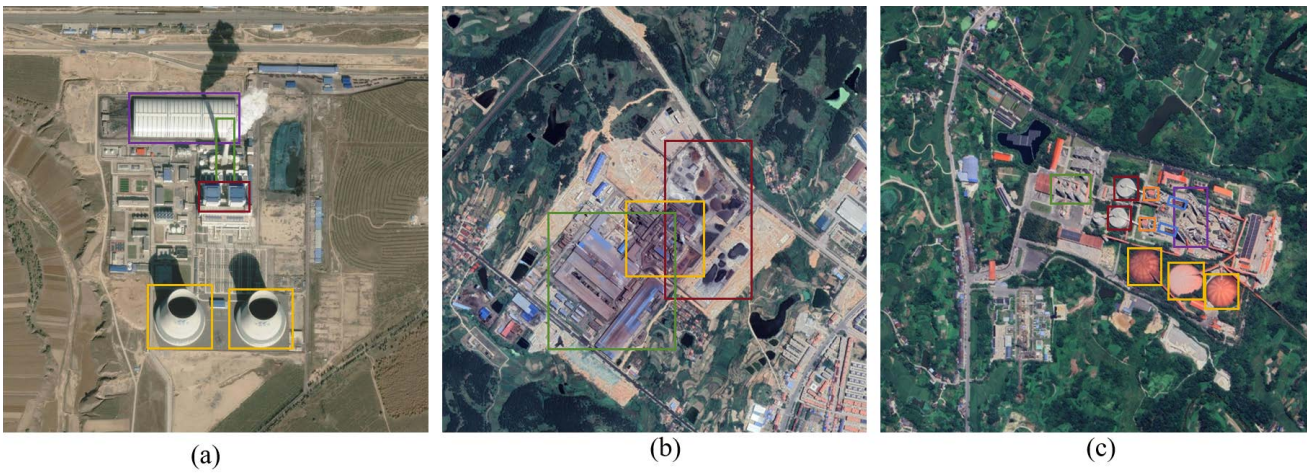


Fig. 1. Industrial composite facilities composed of multiple functional components. (a) TPP with chimney (green bounding box), cooling towers (yellow bounding boxes), boiler building (red bounding box), and turbine hall (purple bounding box). (b) ISP with material yard (red bounding box), steelmaking workshop (green bounding box), and blast furnace and sintering plant (yellow bounding boxes). (c) CP with preheater (purple bounding box), rotary kiln (blue bounding boxes), cooler (orange bounding boxes), clinker reservoir (red bounding boxes), cement silos (green bounding box), and preblending dome (yellow bounding boxes).

confusing substructures, or inaccurate extents) and can be amplified during self-training, causing error accumulation and feature drift. In addition, most SSOD frameworks are not designed to explicitly learn part-level cues, which further reduces robustness in structurally complex scenes. Although transformer-based SSOD methods [20], [21] offer powerful global representation capabilities, accurately identifying composite industrial facilities in RSI still relies heavily on capturing fine-grained part-level evidence.

Detecting industrial facilities from high-resolution RSI has attracted considerable attention. Existing studies have focused on either single-class industrial detection (e.g., fossil-fuel power plants [22], ISPs [23], CPs [24]) or multiclass detection [25], [26], [27], [28]. While these approaches demonstrate the feasibility of large-scale industrial monitoring, they predominantly rely on fully supervised learning and require extensive annotated data, which limits their applicability in regions with scarce annotations. This issue is particularly critical for composite industrial facilities, whose complex internal structures and heterogeneous subcomponents further increase data requirements and sensitivity to supervision noise.

It is worth noting that the practical challenge in industrial monitoring extends beyond annotating individual facilities, but also involves constructing and continuously maintaining large-scale datasets that cover diverse geographic regions, imaging conditions, and temporal variations. Industrial infrastructures are continuously evolving, with newly constructed, expanded, relocated, or decommissioned facilities appearing over time. As a result, obtaining exhaustive annotations for large-scale remote sensing archives requires substantial and recurring manual effort. On the contrary, large volumes of unlabeled RSI are readily available from existing satellite observation platforms. Therefore, SSOD provides a practical and scalable solution by exploiting abundant unlabeled data to improve detection performance while reducing dependence on extensive manual annotations.

Overall, the above discussion highlights three key gaps for industrial composite object detection in RSI: (G1) learning part-aware representations under box-only supervision, (G2) noise accumulation and feature drift in pseudolabel self-training, and (G3) stagewise balancing of pseudolabel quality and quantity.

To alleviate these challenges from a unified perspective, we propose a part-aware multi-dimensional consistency framework (PMC) built upon the Soft Teacher paradigm. Rather than introducing isolated enhancements, PMC aims to improve the effectiveness of pseudosupervised learning through three complementary aspects: discriminative representation learning, reliability-aware pseudosupervision, and adaptive pseudolabel utilization.

In particular, a part-based K-means module (PKM) is designed to discover informative object subregions and strengthen part-aware representations for composite industrial facilities. A multidimensional double-check mechanism (MDCM) enforces complementary structural and frequency consistency constraints to improve pseudosupervision reliability, thereby mitigating noise accumulation and feature drift during self-training. Furthermore, a dynamic top-K pseudolabel selection (DTK) addresses the stagewise imbalance between pseudolabel quality and quantity by dynamically adjusting both confidence thresholding and top-K selection. Together, these components form a coherent learning framework that enhances both representation quality and supervision reliability under limited annotation settings. Unlike existing SSOD methods that mainly optimize individual stages of pseudolabel generation, PMC improves the entire self-training process through three successive stages: representation learning, pseudolabel reliability verification, and adaptive pseudolabel utilization. This design establishes a unified

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learning paradigm rather than an aggregation of isolated module-level improvements.

The main contributions of this article are summarized as follows.

1) A unified semisupervised framework, termed PMC, is proposed for industrial composite object detection in RSI. By jointly improving object representation, pseudosupervision reliability, and pseudolabel utilization, PMC provides a coherent solution for robust self-training under limited annotations.
2) A part-aware representation learning mechanism is introduced to automatically discover discriminative object substructures from pseudolabeled regions, enabling fine-grained structural modeling without requiring additional part-level annotations.
3) A reliability-aware pseudosupervision strategy is developed by integrating multidimensional consistency verification and dynamic curriculum-based pseudolabel selection, which effectively suppresses pseudolabel noise, alleviates feature drift, and stabilizes self-training.
4) Extensive experiments on an industrial composite object dataset and two public benchmarks demonstrate that the proposed framework consistently outperforms fully supervised and state-of-the-art SSOD methods under limited-label scenarios.

The remainder of this article is organized as follows. Section II reviews relevant literature on SSOD, PBMs, and industrial composite object detection. Section III presents the proposed detection framework and the detailed method description. Section IV reports experimental results and analysis. Finally, Section V concludes this article.

II. Related Works

A. Semisupervised Object Detection

Generally, semisupervised learning (SSL) has two categories, closed-set SSL and open-set SSL [29]. Closed-set SSL assumes all unlabeled data belongs to the same set of classes as the labeled data [30], [31], whereas open-set SSL permits the unlabeled data to contain instances from unknown classes or to be mislabeled [32]. In this study, SSL refers specifically to closed-set SSL. Researchers have developed various SSL approaches tailored for object detection, commonly known as SSOD [33].

SSOD has undergone a notable evolution from convolutional neural network (CNN)-based paradigms to transformer-based end-to-end frameworks. CNN-based SSOD methods are generally one-stage or two-stage. One-stage methods, such as DSL [13], OneTeacher [14], and Consistent Teacher [15], focus on stabilizing pseudolabels via consistency regularization, but often struggle with complex backgrounds and small or composite objects. Two-stage methods, such as Soft Teacher [16], PseCo [17], and Active Teacher [18], leverage region proposals to achieve higher localization accuracy and incorporate techniques such as soft pseudolabeling, contrastive learning, and active sample selection. Nevertheless, they still face challenges including pseudolabel bias, domain shifts, and reduced performance on structurally complex industrial objects. Transformer-based SSOD, including Semi-DETR [20] and Omni-DETR [21], provides strong global modeling, but often requires large-scale data and high computational resources, and may inadequately capture fine-grained substructures crucial for composite industrial targets.

SSOD has also been applied to RSI to address high annotation costs and complex scenes [34], [35], [36]. Remote sensing datasets typically present dense, multiscale, and arbitrarily oriented objects with complex backgrounds, challenging conventional SSOD frameworks [37], [38]. Strategies such as cross-branch information incorporation (CBI-SSOD) [35], uncertainty-aware teacher models [39], temporal self-training [40], and dual-teacher frameworks [41] have been proposed to enhance pseudolabel reliability and detection performance.

Despite these advances, composite industrial objects remain challenging due to: 1) amplified pseudolabel noise during self-training and 2) static pseudolabel thresholds or sampling policies that fail to balance label quality and quantity. This motivates developing adaptive pseudolabel selection and more robust consistency regularization for industrial composite object detection.

B. Part-Based Methods

PBMs have long provided a principled framework for addressing nonrigid deformation, pose variation, and occlusion in object detection. The seminal deformable part model (DPM) [10] introduced a root filter with multiple deformable part filters trained via latent SVM, establishing the foundation for explicitly modeling object-part relationships. Subsequent efforts enhanced the robustness and generalizability of DPMs by improving feature stability under deformation [42], enabling unsupervised online boosting for adaptive learning [43] and developing weakly supervised training strategies for learning deformable structures with only image-level labels [44]. PBM concepts were further extended to articulated objects through the articulated PBM (APM), which introduced part types and a coarse-to-fine hierarchical representation for joint detection and pose estimation [45]. In RSI, rotation-invariant PBMs have been explored to handle arbitrary orientations in high-resolution scenes [46].

With the advent of deep learning, PBMs were integrated into CNN-based frameworks. Hybrid approaches combining deformable parts with R-CNN features improved occlusion robustness [47], while deep architectures such as DeePM explicitly learned object-part configurations through a latent graphical model, enabling joint semantic part localization and object detection [11]. In the remote sensing domain, PBNet further demonstrated the utility of part-level cues for complex composite objects such as airports and sewage treatment plants (STPs), introducing explicit part localization and context refinement to handle large structural variations [12]. Other part-based deep models, such as TSPOANet, employed capsule routing to enforce consistent part-object assignments for salient object detection, highlighting the broad applicability of PBM principles.

However, existing PBM-based approaches for remote sensing still depend heavily on dense part annotations or predefined part structures, which often generalize poorly to industrial

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objects exhibiting heterogeneous components and flexible spatial arrangements. Moreover, current methods lack mechanisms to automatically discover fine-grained, discriminative subregions from coarse instance-level labels, making them suboptimal for weakly or semisupervised settings. Motivated by these limitations, this study introduces a simple yet effective part-level representation mechanism for SSOD of complex industrial objects, which encourages the detector to capture meaningful structural cues within pseudolabeled regions and enhances fine-grained discriminative learning without requiring additional part annotations.

C. Industrial Composite Object Detection

Industrial objects such as TPPs, ISPs, and CPs are major sources of energy consumption and carbon emissions [48], [49], [50], [51], [52], [53]. Detecting these objects from high-resolution satellite imagery has, therefore, become an important research topic. Existing studies can be broadly categorized into two groups: single-class industrial object detection and multiclass industrial object detection.

A number of studies focus on detecting one specific type of industrial facility. For example, the BUAA-FFPP60 dataset was introduced for fossil fuel power plant detection and operational state recognition, where eight representative deep learning detectors were benchmarked on high-resolution imagery [22]. Similarly, several works address steel plant monitoring by exploiting thermal infrared signals. The iron and steel heat island intensity index (ISHII) and heat ratio index (ISHRI) were proposed to characterize the operational dynamics of ISPs based on Landsat-8 TIRS and GF-2 imagery, enabling differentiation between production, shutdown, and curtailment states [23]. Follow-up studies further applied seasonal-trend decomposition to improve the temporal stability of steel-plant activity indicators [54]. In the cement industry, global CP mapping has been conducted using multisource Landsat-8 and Sentinel-2 imagery, supported by semiautomatic kiln recognition [24].

Beyond single-class studies, several works have explored the detection of multiple categories of heavy industrial objects from RSI. Early work used high-resolution optical imagery to detect ISPs, CPs, and chemical plants [25]. Thermal anomaly data and SDGSAT-1 TIS observations further enable national-scale and small-scale detection [26], [27]. Recent datasets such as the multiclass industrial heat source (IHS) dataset, combined with points of interest (POI) information, support heterogeneous object detection with frameworks like YOLOv8-FC incorporating FasterNet and coordinate attention [28].

Existing industrial object detection methods still suffer from several limitations. First, high-quality industrial object datasets are difficult to obtain, as most existing benchmarks rely on fully supervised annotations, which are labor-intensive and limit scalability. Second, industrial facilities often exhibit complex and heterogeneous structures (e.g., chimneys, boilers, and kilns), making it difficult to learn discriminative representations with limited labeled samples. Third, thermal anomaly data are commonly used for industrial object screening, but they mainly support coarse identification and still require manual refinement for fine-grained categorization. These limitations motivate the exploration of semisupervised industrial object detection methods that can leverage abundant unlabeled imagery for scalable and accurate monitoring.

III. Methodology

Our proposed PMC framework aims to enhance semisupervised detection of industrial objects through part-aware representation learning, reliability-aware pseudosupervision, and adaptive pseudolabel utilization. Different from conventional SSOD frameworks that mainly rely on confidence-based pseudolabel filtering, PMC introduces a unified learning paradigm that progressively improves pseudosupervision from three interconnected stages. As illustrated in Fig. 2, the overall methodology consists of three key components: PKM, MDCM, and DTK. In particular, PKM enhances object representation through auxiliary pseudosupervision, MDCM evaluates pseudolabel reliability through consistency constraints, and DTK adaptively controls pseudolabel selection and utilization during self-training. These components are seamlessly integrated into the Soft Teacher paradigm to extend conventional confidence-based pseudolabel learning toward a more reliable and adaptive self-training framework for industrial composite object detection.

A. Part-Based K-Means Module

To enhance the model's ability to learn fine-grained discriminative representations under limited supervision, we design a PKM, which automatically discovers discriminative local regions within pseudolabeled regions through feature clustering. This module encourages the network to focus on discriminative visual regions of industrial objects, such as characteristic areas in cooling towers, chimneys, and tanks, thereby improving robustness against background interference and intraclass variation.

Given the pseudoground truth instances generated by the teacher network, the PKM module further mines several local regions within each coarse bounding box without requiring additional annotations. In particular, for each detected instance, we extract its corresponding feature patch from the teacher feature map and apply unsupervised clustering to identify representative subregions. The detailed procedure is presented in Algorithm 1.

1) Feature Cropping: Unlike global feature clustering, PKM performs clustering only inside each pseudo object region. This object-level constraint preserves the spatial prior provided by the teacher predictions and prevents irrelevant background regions from participating in clustering. Meanwhile, the centroid coordinates of clustered features retain the spatial arrangement information inside each object region. Let $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ denote the feature map of an image, and $\mathbf{b} = [x_1, y_1, x_2, y_2]$ be a bounding box predicted by the teacher model. The corresponding feature patch $\mathbf{F_b}$ is obtained by cropping $\mathbf{F}$ within the region of $\mathbf{b}$

$$\mathbf{F_b} = \text{Crop}(\mathbf{F}, \mathbf{b}), \quad \mathbf{F_b} \in \mathbb{R}^{C\times h\times w}. \tag{1}$$

Each pixel in $\mathbf{F_b}$ is treated as a C-dimensional feature vector representing local spatial semantic cues.

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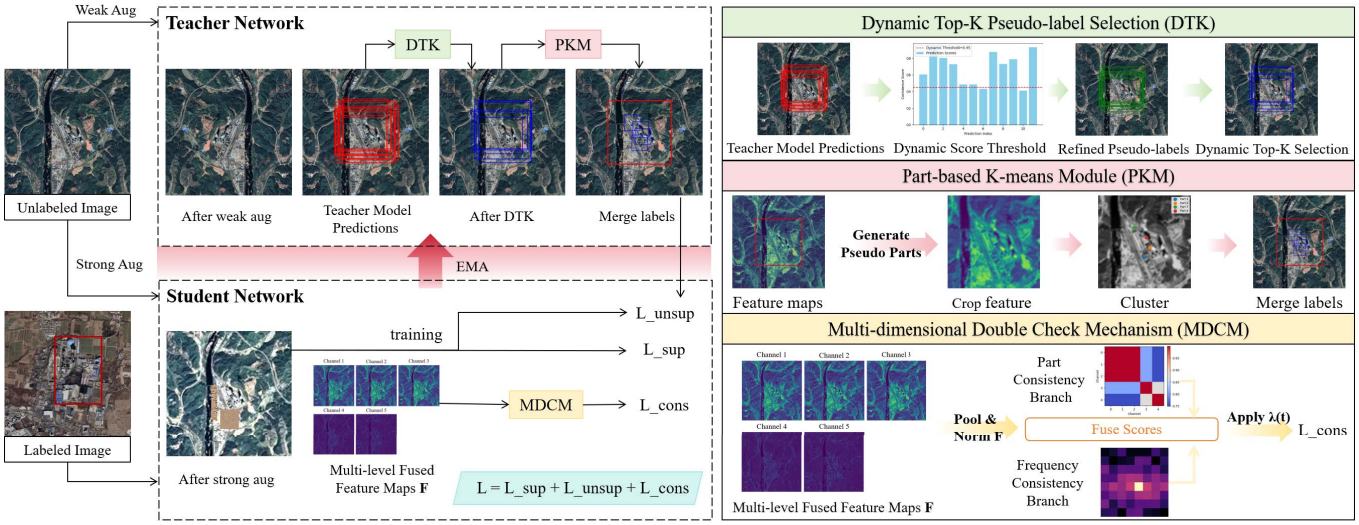


Fig. 2. Overview of the proposed semisupervised detection framework.

Algorithm 1 PKM

Require: Feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, pseudobounding boxes $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_N\}$ from the teacher model, number of clusters K

Ensure: Augmented pseudobounding box set $\mathcal{B}'$

Initialize $\mathcal{B}' \leftarrow \emptyset$

for each bounding box $\mathbf{b} = [x_1, y_1, x_2, y_2] \in \mathcal{B}$ **do**

Feature Cropping: Extract the corresponding feature patch

$$\mathbf{F}_\mathbf{b} = \text{Crop}(\mathbf{F}, \mathbf{b}), \quad \mathbf{F}_\mathbf{b} \in \mathbb{R}^{C\times h\times w}$$

Reshape $\mathbf{F}_\mathbf{b}$ into a set of feature vectors $\{\mathbf{f}_i\}_{i=1}^{hw}$

K-means Clustering: Apply K-means to obtain K part centroids $\{\mu_k\}_{k=1}^{K}$:

$$\min_{\{\mu_k\}} \sum_{i=1}^{hw} \min_k \|\mathbf{f}_i - \mu_k\|^2$$

For each cluster k, compute centroid position (c_x^k, c_y^k) and map it back to image coordinates using stride s

Construct part bounding box:

$$\mathbf{p}_k = [c_x^k - w_p/2, c_y^k - h_p/2, c_x^k + w_p/2, c_y^k + h_p/2]$$

where $w_p = (x_2 - x_1)/4$ and $h_p = (y_2 - y_1)/4$

Collect $\mathcal{B}_{\text{parts}} = \{\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_K\}$

Part Integration: Merge part boxes with object box:

$$\mathcal{B}' \leftarrow \mathcal{B}' \cup \{\mathbf{b}\} \cup \mathcal{B}_{\text{parts}}$$

end for

return $\mathcal{B}'$

2) K-Means-Based Part Localization: The cropped features are reshaped into a set of vectors $\{\mathbf{f}_i\}_{i=1}^{hw}$, which are then clustered using K-means with K clusters

$$\min_{\{\mu_k\}} \sum_{i=1}^{hw} \min_k \|\mathbf{f}_i - \mu_k\|^2 \tag{2}$$

where μ_k denotes the centroid of the kth part cluster. Each cluster represents a feature-consistent local region that contains potentially discriminative visual patterns within the object instance. The centroid position (c_x^k, c_y^k) within the feature map is then mapped back to the image coordinate system according to the feature stride s, and a part bounding box $\mathbf{p}_k = [c_x^k - w_p/2, c_y^k - h_p/2, c_x^k + w_p/2, c_y^k + h_p/2]$ is constructed, where $w_p = (x_2 - x_1)/4$ and $h_p = (y_2 - y_1)/4$.

3) Part Integration: For each pseudolabeled instance, the generated part boxes and the original coarse bounding box are merged to form an augmented pseudoset

$$\mathcal{B}' = \mathcal{B}_{\text{object}} \cup \mathcal{B}_{\text{parts}} \tag{3}$$

where $\mathcal{B}_{\text{object}}$ and $\mathcal{B}_{\text{parts}}$ denote the bounding boxes of the original objects and the discovered parts, respectively. This integration encourages the detector to attend to multiple discriminative subregions within each object, thereby facilitating part-aware representation learning and capturing complementary local visual patterns.

B. Multidimensional Double Check Mechanism

To mitigate error accumulation caused by noisy pseudolabels, we introduce an MDCM to improve the consistency and reliability of pseudosupervision. MDCM evaluates pseudosupervision reliability from two complementary representation perspectives.

1) A structural consistency score that measures the coherence of correlated local structural responses encoded by feature channels.
2) A spectral regularity score that encourages stable frequency-domain representations and alleviates response instability caused by noisy pseudosupervision. By jointly constraining structural coherence and spectral regularity, MDCM provides a cross-dimensional regularization signal that suppresses unreliable pseudosupervision and stabilizes self-training.

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1) Structural Consistency Score: Let $\mathbf{F} \in \mathbb{R}^{B\times C\times H\times W}$ denote the multilevel fused feature maps. To reduce computational cost, $\mathbf{F}$ is adaptively pooled to a fixed spatial size $(8, 8)$

$$\mathbf{F}' = \text{Pool}\left(\mathbf{F}, 8 \times 8\right). \tag{4}$$

Each channel feature map is then normalized along the spatial dimension

$$\tilde{\mathbf{F}} = \text{Norm}\left(\mathbf{F}' - \text{Mean}\left(\mathbf{F}'\right)\right). \tag{5}$$

Although individual feature channels do not correspond to explicit semantic parts, convolutional filters generally capture diverse local visual patterns, and channels encoding related structural characteristics may exhibit correlated activation behaviors. For industrial composite facilities, these local structures frequently recur across multiple functional components. When pseudosupervision is reliable, channels encoding related structural patterns tend to exhibit coherent activations, whereas noisy pseudolabels usually introduce unstable responses and weaken such correlations. Therefore, channelwise similarity is employed as an implicit indicator of local structural consistency rather than explicit part semantics.

The pairwise channel similarity is computed as

$$\mathbf{S} = \tilde{\mathbf{F}}\tilde{\mathbf{F}}^{\top}, \quad \mathbf{S} \in \mathbb{R}^{B\times C\times C}. \tag{6}$$

Local channelwise similarity is adopted as a lightweight surrogate for local channel correlation patterns while avoiding the computational cost of evaluating all channel pairs. Instead of averaging all similarities, only adjacent-channel similarities are considered to characterize local channel correlation patterns associated with structural consistency. The structural consistency score is computed as

$$\text{Score}_{\text{str}} = \frac{1}{B\left(C-1\right)} \sum_{b=1}^{B} \sum_{c=1}^{C-1} \mathbf{S}_{b,c,c+1}. \tag{7}$$

A higher value of $\text{Score}_{\text{str}}$ indicates stronger correlations among channels encoding related local structures, suggesting that the learned representation preserves more stable structural patterns and is less affected by noisy pseudosupervision.

2) Spectral Regularity Score: To further enhance the stability of frequency-domain representations, the pooled features are transformed using a 2-D fast Fourier transform (FFT)

$$\mathbf{M} = \text{FFTShift}\left(\text{FFT2}\left(\text{Mean}_C\left(\mathbf{F}'\right)\right)\right) \tag{8}$$

where $\mathbf{M}$ represents the centered frequency magnitude. The normalized magnitude spectrum is given by

$$\tilde{\mathbf{M}} = \frac{|\mathbf{M}| - \mu_{\mathbf{M}}}{\sigma_{\mathbf{M}} + \epsilon}. \tag{9}$$

The spectral regularity score is computed as

$$\text{Score}_{\text{spec}} = \frac{1}{B} \sum_{b=1}^{B} \frac{\left\langle \tilde{\mathbf{M}}_b, \tilde{\mathbf{M}}_b \right\rangle}{\left\| \tilde{\mathbf{M}}_b \right\|_2^2 + \epsilon}. \tag{10}$$

Industrial facilities usually exhibit relatively regular spatial layouts and repetitive structural patterns. Such characteristics tend to produce stable frequency responses in the learned feature space. On the contrary, noisy pseudosupervision may introduce unstable activations and abnormal spectral variations. By regularizing the normalized frequency-domain responses, the proposed score helps maintain stable spectral representations and reduces the influence of noisy pseudosupervision during self-training. The structural and spectral criteria characterize representation reliability from complementary perspectives. The former focuses on local structural coherence, while the latter regularizes the magnitude distribution of frequency-domain responses. Although a simple averaging strategy is adopted, the two scores exhibit different sensitivities to pseudolabel noise and therefore contribute distinct regularization effects. Considering the lightweight role of MDCM as an auxiliary regularizer, a simple averaging strategy is intentionally adopted to avoid introducing additional learnable parameters and optimization instability.

Finally, both scores are combined with a decay weighting strategy to form the overall multidimensional consistency loss

$$\mathcal{L}_{\text{cons}} = \lambda\left(t\right)\left[1 - \frac{1}{2}\left(\text{Score}_{\text{str}} + \text{Score}_{\text{spec}}\right)\right] \tag{11}$$

where $\lambda(t) = \max(0.1, 1 - t/T_{\max})$ is an iterative decay coefficient to gradually reduce the regularization strength as training progresses.

C. Dynamic Top-K Pseudolabel Selection

To improve the reliability of pseudolabels generated by the teacher model, we propose a DTK strategy. DTK combines two coupled operations: a dynamic score threshold and a dynamic top-K selection, to adaptively balance pseudolabel coverage and reliability during semisupervised training.

1) Dynamic Score Threshold: To progressively relax the selection criteria during training, the score threshold is linearly decayed over iterations

$$\tau_{\text{dyn}} = \max\left(\tau_{\min}, \tau_{\text{base}} \cdot \left(1 - t/T_{\max}\right)\right) \tag{12}$$

where t is the current iteration, $T_{\max}$ is the maximum number of training iterations, τ_{base} is the initial threshold, and $\tau_{\min}$ is the minimum allowed threshold. This allows the early-stage selection to be conservative, while gradually including more pseudolabels as training progresses.

2) Dynamic Top-K Selection: In addition to thresholding, a top-K strategy is applied to select the most confident pseudoinstances for each image. The number of selected instances $K(t)$ grows linearly over time

$$K\left(t\right) = K_{\min} + \left(K_{\max} - K_{\min}\right) \cdot t/T_{\max} \tag{13}$$

where $K_{\min}$ and $K_{\max}$ denote the minimum and maximum allowed pseudoinstances per image. Only the $K(t)$ highest scoring pseudoinstances that exceed τ_{dyn} are kept, ensuring a balance between pseudolabel reliability and coverage.

These modifications are applied after the teacher model predicts bounding boxes and scores, prior to any uncertainty computation or projection to the original image coordinates. Unless otherwise specified, the DTK parameters are fixed as $\tau_{\text{base}} = 0.3$, $\tau_{\min} = 0.05$, $K_{\min} = 20$, and $K_{\max} = 200$ throughout all experiments. Here, $K_{\max}$ serves as an upper bound on the number of retained pseudolabels rather than a fixed selection number, and the actual number of selected

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pseudolabels is jointly determined by the dynamic threshold and confidence scores. Overall, the proposed DTK mechanism adaptively balances pseudolabel reliability and diversity during different training stages, enabling more effective utilization of pseudolabels for semisupervised self-training.

D. Loss Function

Our framework integrates multibranch supervision under semisupervised settings, combining supervised learning on labeled data with reliability-aware pseudosupervision on unlabeled data. The overall optimization is jointly formulated through supervised detection loss, reliability-aware pseudosupervision loss, and multidimensional consistency regularization, which correspond to part-aware representation learning, reliable pseudolabel verification, and adaptive pseudolabel utilization, respectively.

1) Supervised Loss: For labeled images, the standard detection loss is applied to the ground truth instances

$$\mathcal{L}_{\text{sup}} = \text{Loss}_{\text{GT}}\left(\mathbf{X}_{\text{sup}}, \mathcal{B}_{\text{sup}}\right) \tag{14}$$

where $\mathbf{X}_{\text{sup}}$ denotes supervised input images and $\mathcal{B}_{\text{sup}}$ the corresponding ground truth bounding boxes and labels.

2) Pseudolabel Loss: For unlabeled images, pseudoinstances are generated by the teacher model with the *DTK*. These pseudoinstances are further enriched with discriminative local regions generated by the PKM. These local pseudoboxes are incorporated together with the original object-level pseudoboxes to provide complementary supervision

$$\mathcal{B}'_{\text{unsup}} = \mathcal{B}_{\text{pseudo}} \cup \mathcal{B}_{\text{parts}}. \tag{15}$$

The student model is then trained on these augmented pseudolabels

$$\mathcal{L}_{\text{unsup}} = \text{Loss}_{\text{pseudo}}\left(\mathbf{X}_{\text{unsup}}, \mathcal{B}'_{\text{unsup}}\right). \tag{16}$$

It is worth noting that the PKM-generated local pseudoboxes are not considered as independent object instances or strict semantic object parts. Instead, they represent discriminative subregions within the corresponding pseudo object and inherit the category-level label information of the parent instance. These local regions serve as auxiliary supervision for learning part-aware representations, while the original pseudobounding boxes remain the primary supervision for object-level localization. By providing complementary local cues within the same composite object region, PKM enriches the supervision signal and mitigates potential semantic ambiguity between local regions and the original pseudolabels.

3) Multidimensional Consistency Loss: To regularize the student network and suppress noisy pseudosupervision, we enforce structural coherence and spectral regularity, as described in Section III-B

$$\mathcal{L}_{\text{cons}} = \lambda(t)\left[1 - \frac{1}{2}\left(\text{Score}_{\text{str}} + \text{Score}_{\text{spec}}\right)\right] \tag{17}$$

where $\lambda(t)$ is a decay coefficient that gradually reduces the strength of the regularization over training iterations.

4) Overall Loss: The total loss for a mini-batch combining supervised and unlabeled branches is

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \mathcal{L}_{\text{unsup}} + \mathcal{L}_{\text{cons}}. \tag{18}$$

This design enables effective SSL by leveraging part-aware representations, reliable pseudosupervision, and adaptive pseudolabel utilization for robust self-training.

IV. Experimental Results and Analysis

A. Datasets and Evaluation Metrics

1) Datasets: In this study, an industrial composite object image dataset (Indus-CO) was constructed using POI information and level-17 high-resolution imagery from Google Earth. After careful selection and verification, a total of 2090 image samples were collected, including 950 from TPPs, 791 from CPs, and 349 from ISPs. Each image covers an actual ground area of approximately 4096 $\times$ 4096 m and is cropped with the factory region as the center to ensure target integrity and contextual completeness.

During the annotation phase, the research team adopted the COCO format to ensure consistency across different categories, providing horizontal bounding boxes and corresponding class labels for each factory instance. The dataset spans various geomorphological types in mainland China, including plains, coastal zones, and hilly regions, thereby exhibiting substantial variation in land cover, background composition, and structural complexity. This diversity provides a solid foundation for evaluating industrial composite object detection under complex environmental conditions.

To further validate the generalization capability of the proposed model, two composite datasets were derived from publicly available benchmarks. Specifically, based on the DIOR dataset [55], a subset named *DIOR-composite* was created by selecting multifeature targets, including *airport (APR)*, *harbor (HB)*, and *expressway service area (ESA)*, yielding 3893 images in total. Similarly, from the DOTA-v2.0 dataset [56], the *DOTA-composite* subset was constructed using *airport* and *harbor* categories, resulting in 914 images. These composite datasets effectively extend the evaluation scope to different domains and object types, providing a more comprehensive assessment of the robustness and adaptability of the model.

Besides, another industrial composite object dataset named *Indus-GF* was constructed to evaluate the generalization performance on imagery acquired by Chinese Earth observation satellites. The Indus-GF dataset comprises 1304 images of TPPs and STPs, collected from the Gaofen-2 and Gaofen-6 satellites. By incorporating multisource satellite data with distinct imaging characteristics and spectral responses, Indus-GF serves as an additional testbed for examining the adaptability of the proposed method to high-resolution Chinese satellite imagery, thereby providing a practical assessment of its deployment potential across different Earth observation platforms. The detailed composition statistics of the newly constructed industrial composite object datasets are summarized in Table I.

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TABLE I
COMPOSITION OF THE NEWLY CONSTRUCTED INDUSTRIAL COMPOSITE OBJECT DATASETS

Dataset	Class Category	Number of Images
Indus-CO	Thermal Power Plant (TPP)	950
	Cement Plant (CP)	791
	Iron and Steel Plant (ISP)	349
	Total	**2,090**
Indus-GF	Thermal Power Plant (TPP)	481
	Sewage Treatment Plant (STP)	823
	Total	**1,304**

2) Data Partition: For the SSL experiments, we first construct fixed validation and test splits to ensure unbiased evaluation. In particular, 20% of the total dataset is held out as the validation set, while the remaining images form the training pool used for SSL. Within this training pool, three labeled data ratios are considered (10%, 25%, and 50%): for each ratio, the labeled subset is randomly sampled in a class-balanced manner, and the remaining training images are treated as unlabeled data.

3) Evaluation Metrics: Consistent with common practice in remote sensing object detection, model performance was assessed using the mean average precision (mAP). In particular, both mAP_{50} and $mAP_{50:95}$ were adopted for evaluation. mAP_{50} denotes the average precision at an intersection-over-union (IoU) threshold of 0.5, while $mAP_{50:95}$ represents the averaged AP across multiple IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05, following the COCO evaluation protocol. By jointly reporting mAP_{50} and $mAP_{50:95}$, the proposed method can be more comprehensively evaluated in terms of localization accuracy and detection robustness under different supervision settings.

B. Implementation Details

All experiments are implemented based on the MMDetection framework. We adopt a Soft Teacher [16] semisupervised detection framework with a Faster R-CNN [57] detector and a ResNet-50 backbone with FPN [58]. Training follows an iteration-based loop with a total of 48k iterations, and validation is performed every 2k iterations. The optimizer is AdamW with a learning rate of 0.0002 and weight decay of 0.0001. A two-stage learning rate schedule is used: a linear warm-up for the first 1k iterations, followed by multistep decay at 32k and 40k iterations. All experiments are conducted on a single NVIDIA A5000 GPU. During inference, predictions are made using the teacher model. For fair comparison, the supervised baseline was implemented using the same Faster R-CNN detector, ResNet-50-FPN backbone, optimizer settings, and training schedule as the Soft Teacher framework, while removing the unlabeled data branch and training only with labeled data. Our code and dataset are available at https://github.com/Gabrielle-He/SSOD-PMC

C. Comparisons With the SOTA

This section presents a comprehensive comparison between the proposed method PMC and several representative state-of-the-art SSOD approaches on the Indus-CO dataset. The compared methods include Soft Teacher, DCST, DSL, SSOD-AT, Active Teacher, Semi-DETR, Unbiased Teacher [59], Unbiased Teacher v2 [60], and Consistent Teacher, which cover a broad range of mainstream paradigms. For a fair comparison, all methods are evaluated under the same experimental settings and data splits, following the standard evaluation protocol of the Indus-CO dataset. In particular, the supervised baseline adopts the same Faster R-CNN architecture and training configuration as Soft Teacher, differing only in the absence of unlabeled data and teacher–student learning components.

As summarized in Table II, the proposed PMC consistently achieves superior or highly competitive performance across different labeled ratios. Under the low-label regime (10%), PMC attains the highest category-average mAP_{50} of 64.1% and $mAP_{50:95}$ of 30.2%, outperforming all compared semisupervised baselines. In particular, PMC shows clear advantages on the ISP category, achieving 45.3% mAP_{50}, indicating its strong capability in modeling complex industrial facilities under limited supervision. When the labeled ratio increases to 25%, PMC further improves its performance and achieves the best overall mAP_{50} of 71.8% and $mAP_{50:95}$ of 35.8%. Compared with representative teacher–student-based methods such as Soft Teacher and DCST, the proposed method demonstrates consistent gains, especially on ISP detection, where PMC reaches 57.9% mAP_{50}. This suggests that PMC can effectively exploit unlabeled data to learn more discriminative and structure-aware representations. Under the 50% labeled setting, PMC continues to achieve the best overall performance, obtaining 75.8% mAP_{50} and 38.9% $mAP_{50:95}$. Although the improvement in mAP_{50} becomes relatively smaller as more labeled data are available, the gain in $mAP_{50:95}$ remains notable. Compared with Soft Teacher, PMC improves $mAP_{50:95}$ from 35.8% to 38.9%, demonstrating more accurate localization under stricter IoU thresholds. These results indicate that PMC not only improves detection accuracy but also enhances localization robustness by producing more reliable pseudosupervision across different annotation ratios.

In addition to quantitative comparisons, Fig. 3 presents qualitative detection results under the 25% labeled setting. The proposed method produces more accurate and complete object localization for industrial composite facilities, particularly for objects with complex internal structures. Compared with existing methods, PMC yields fewer false positives and tighter bounding boxes, which is consistent with the numerical improvements reported in Table II. These visual results further confirm the effectiveness of the proposed approach.

D. Ablation Studies

To thoroughly assess the effectiveness of each proposed component, we conduct comprehensive ablation studies on three key modules of our framework: PKM, MDCM, and DTK. All experiments are performed on the self-constructed Indus-CO dataset under the 25% labeled setting, while keeping all other training configurations identical to ensure a fair comparison. This 25% labeled protocol reflects a typical SSOD regime where annotations are scarce and pseudolabel noise is

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TABLE II

COMPARISON OF SEMISUPERVISED METHODS UNDER DIFFERENT LABELED RATIOS ON THE INDUS-CO DATASET. THE SUPERVISED BASELINE IS IMPLEMENTED USING THE SAME FASTER R-CNN AND RESNET-50-FPN CONFIGURATION AS SOFT TEACHER UNDER IDENTICAL TRAINING SCHEDULES

Method	10%					25%					50%				
	TPP	CP	ISP	mAP_{50}	$mAP_{50:95}$	TPP	CP	ISP	mAP_{50}	$mAP_{50:95}$	TPP	CP	ISP	mAP_{50}	$mAP_{50:95}$
Supervised	53.6	58.7	29.6	47.3	18.5	67.8	72.1	40.6	60.2	28.2	73.1	74.9	49.4	65.8	33.0
DSL [13]	72.3	**75.1**	43.1	63.5	28.5	77.3	**82.8**	50.0	70.0	32.2	81.0	82.3	52.5	72.0	34.5
Active Teacher [18]	64.0	56.4	38.7	53.0	19.5	73.9	65.6	38.0	59.2	25.3	73.1	73.6	54.1	66.9	30.4
DCST [61]	**75.7**	73.0	35.6	61.4	27.9	**79.4**	77.1	48.6	68.4	32.7	**84.2**	81.3	59.3	75.0	36.6
Soft Teacher [16]	73.2	72.7	42.1	62.7	28.6	78.7	74.3	54.5	69.2	34.9	82.6	80.9	58.3	73.9	35.8
SSOD-AT [38]	62.6	54.0	37.6	51.4	20.3	72.8	64.3	36.1	57.7	24.9	74.2	70.9	50.8	65.3	30.1
Unbiased Teacher [59]	64.7	68.7	32.0	55.1	23.7	80.0	76.8	40.8	65.9	32.3	79.5	77.0	65.0	73.9	37.0
Unbiased Teacher v2 [60]	66.4	63.4	37.1	55.6	24.8	73.1	77.9	53.4	68.1	33.3	80.3	79.2	65.5	75.0	37.9
Consistent-Teacher [15]	36.6	64.2	22.8	41.2	16.7	49.2	55.1	28.6	44.3	17.6	56.7	63.4	29.3	49.8	19.3
Semi-DETR [20]	14.1	13.2	12.8	13.4	7.0	12.7	14.7	14.8	14.1	8.0	16.3	18.3	19.8	18.1	11.0
PMC (Ours)	73.8	73.4	**45.3**	**64.1**	**30.2**	78.6	79.0	**57.9**	**71.8**	**35.8**	83.5	**82.6**	**61.2**	**75.8**	**38.9**

TABLE III

EFFECT OF THE PKM

Method	mAP_{50} (%)	Gain
Soft Teacher (baseline)	69.2	-
Soft Teacher + PKM	70.4	+1.2
PMC (Ours)	71.8	+2.6

TABLE IV

INTERNAL ABLATION OF THE MDCM UNDER THE COMPLETE PMC FRAMEWORK. S = STRUCTURAL CONSISTENCY SCORE; R = SPECTRAL REGULARITY SCORE

Configuration	S	R	mAP_{50} (%)	Gain
PMC w/o MDCM	-	-	70.7	-
PMC w/o R (S only)	✓	-	71.3	+0.6
PMC w/o S (R only)	-	✓	71.2	+0.5
PMC (S + R)	✓	✓	71.8	+1.1

TABLE V

EFFECT OF THE DTK

Method	mAP_{50} (%)	Gain
Soft Teacher (baseline)	69.2	-
Soft Teacher + DTK	69.8	+0.6
PMC (Ours)	71.8	+2.6

more pronounced, making the contributions of PKM, MDCM, and DTK easier to observe and distinguish.

1) Effect of the PKM: Table III summarizes the impact of PKM on detection performance. Starting from the Soft Teacher baseline, introducing PKM improves the mAP from 69.2% to 70.4%, yielding a gain of + 1.2%. This result demonstrates that PKM effectively enhances part-aware feature representation by automatically discovering discriminative local regions within coarse pseudolabeled bounding boxes, without relying on explicit part-level annotations.

By encouraging the network to attend to structurally meaningful subcomponents of industrial composite objects, PKM provides more informative and reliable supervision signals during pseudolabel refinement. When combined with the other proposed modules, the full model further achieves an mAP of 71.8%, indicating that PKM plays a complementary role within the overall framework rather than acting as an isolated performance booster.

2) Internal Ablation of the MDCM: To better verify the individual contributions of the two consistency criteria in MDCM, we conduct controlled ablation experiments under the complete PMC framework while selectively disabling the structural consistency score (S) and spectral regularity score (R). Table IV summarizes the corresponding results. When both consistency constraints are removed, the framework achieves a baseline performance of 70.7%. Introducing only the S improves the mAP_{50} to 71.3%, while using only the R yields 71.2%. These results demonstrate that both structural coherence and spectral regularity contribute positively to pseudolabel reliability. Moreover, combining the two criteria further boosts the performance to 71.8%, outperforming either individual component. This observation suggests that the two consistency measures provide complementary supervision cues from different representation perspectives. In particular, the S focuses on preserving stable local structural responses, whereas the R captures global spectral stability and reduces the influence of unreliable pseudosupervision with unstable frequency responses. By jointly enforcing these complementary constraints, MDCM effectively improves pseudolabel quality and stabilizes the semisupervised training process.

3) Effect of the DTK: The effect of the proposed DTK strategy is reported in Table V. Compared with the Soft Teacher baseline, dynamically adjusting the number of selected pseudolabels improves the mAP from 69.2% to 69.8%, corresponding to a gain of + 0.6%. Although the standalone improvement is relatively modest, this result suggests that DTK can better adapt to varying confidence distributions among unlabeled samples, thereby reducing the negative impact of low-quality predictions. More importantly, DTK plays a critical role when combined with PKM and MDCM. By dynamically balancing pseudolabel quantity and quality throughout training, DTK enhances the robustness of the SSL process, ultimately contributing to the best overall performance achieved by the full model.

4) Overall Module Combination Analysis: Table VI provides a comprehensive comparison of different module combinations. Each individual component consistently improves

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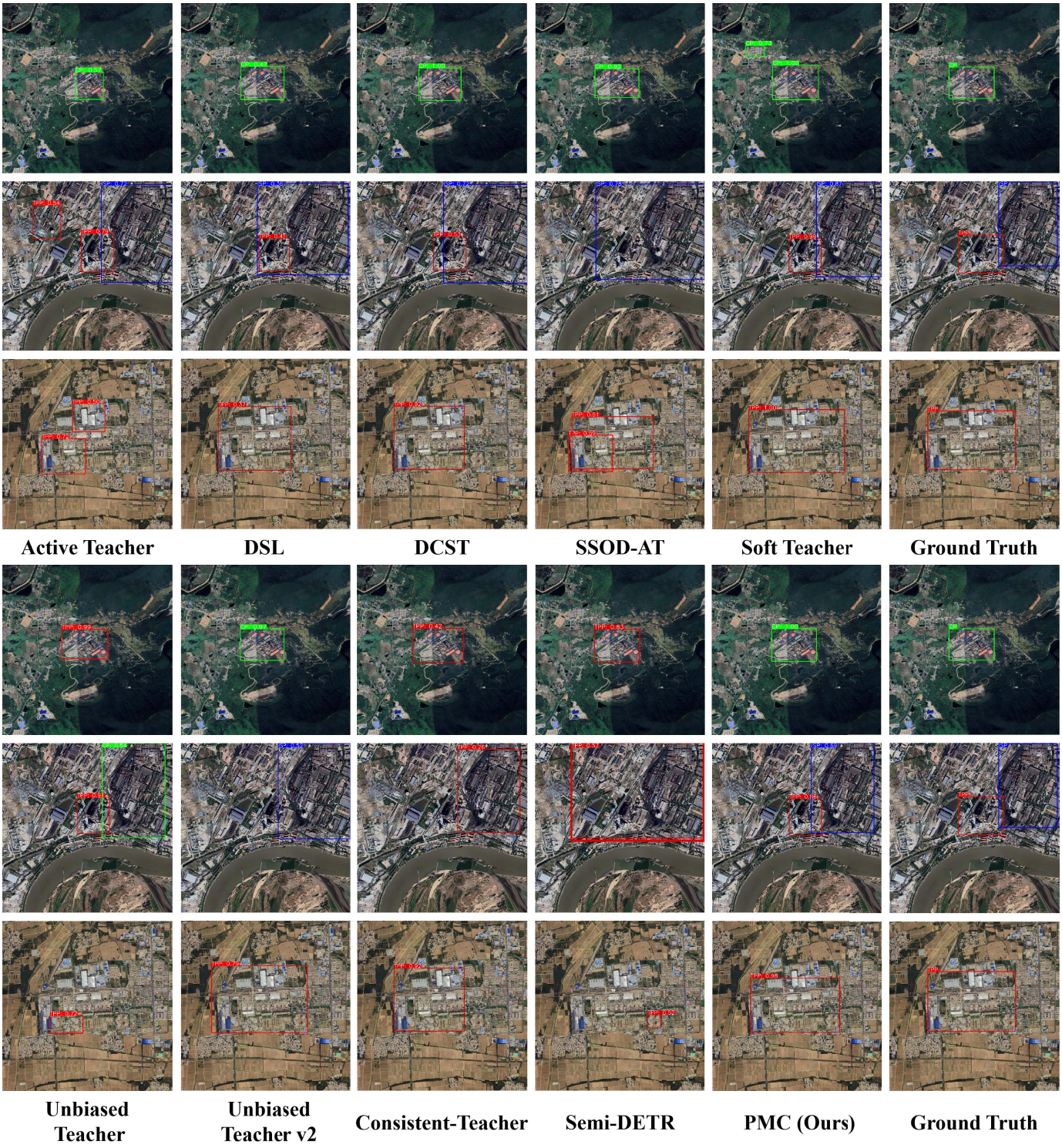


Fig. 3. Qualitative comparison of detection results under a 25% labeled setting between the proposed method and representative state-of-the-art semisupervised approaches on the Indus-CO dataset. (TPP, red bounding boxes; ISP, blue bounding boxes; and CP, green bounding boxes).

performance over the baseline, while combining multiple modules leads to further gains. In particular, configurations that jointly exploit PKM and MDCM exhibit more robust detection performance. When all three modules are enabled, the proposed framework achieves the highest mAP of 71.8%, clearly demonstrating the strong complementarity among PKM, MDCM, and DTK.

5) Training Dynamics Analysis: Fig. 4 illustrates the training dynamics in terms of mAP over iterations for different configurations, including the baseline, baseline + PKM, baseline + MDCM, baseline + DTK, and the full model. The trends observed in the curves are consistent with the quantitative results reported in Tables III–VI. A noticeable performance drop around 32k iterations is observed across all

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TABLE VI
OVERALL ABLATION STUDY ON THE EFFECTIVENESS OF DIFFERENT MODULE COMBINATIONS

Configuration	PKM	MDCM	DTK	mAP_{50} (%)
Soft Teacher (baseline)	-	-	-	69.2
+ PKM	✓	-	-	70.4
+ MDCM	-	✓	-	70.6
+ DTK	-	-	✓	69.8
+ PKM + MDCM	✓	✓	-	71.3
+ PKM + DTK	✓	-	✓	70.7
+ MDCM + DTK	-	✓	✓	71.0
PMC (Ours)	✓	✓	✓	71.8

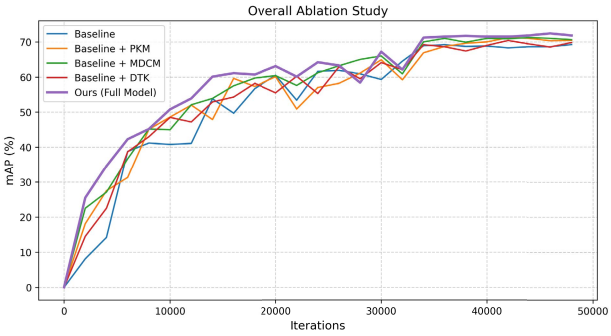


Fig. 4. Training curves of mAP versus training iterations for different configurations, including the Soft Teacher baseline, baseline with PKM, baseline with MDCM, baseline with DTK, and the proposed full model on the Indus-CO dataset under the 25% labeled setting. (TPP, red bounding boxes; ISP, blue bounding boxes; and CP, green bounding boxes).

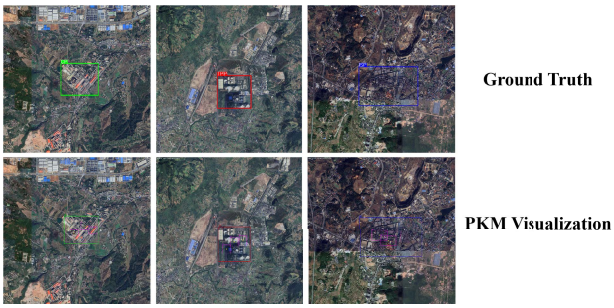


Fig. 5. Visualization of the local regions generated by PKM.

configurations, which is mainly caused by the scheduled learning rate decay that temporarily alters the optimization dynamics and leads to a short-term drop in mAP.

E. Feature Visualization and Interpretability Analysis

1) Visualization of PKM: Fig. 5 visualizes the local regions automatically discovered by the proposed PKM module. It should be noted that these regions are not intended to correspond to true semantic components (e.g., cooling towers or chimneys) of composite industrial objects. Instead, PKM generates feature-consistent local regions with predefined relative sizes, which serve as auxiliary supervision during training. The objective is to encourage the detector to attend to multiple discriminative subregions within each object and improve part-aware representation learning under limited annotations, rather than performing explicit semantic part decomposition.

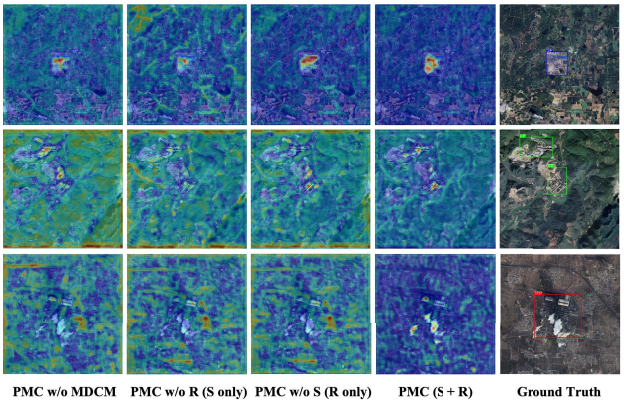


Fig. 6. Feature-response visualization of MDCM. The heatmaps are generated from the most activated feature channels of the high-level feature maps. Warmer colors indicate stronger feature responses.

Therefore, PKM should be regarded as a weakly supervised local region mining strategy rather than a semantic part discovery method.

2) Visualization of MDCM: To further understand the effect of MDCM, Fig. 6 visualizes the high-level feature responses after applying the proposed consistency regularization. The highlighted regions mainly concentrate on the dominant structural components of industrial facilities rather than background areas. Moreover, the activated responses exhibit coherent spatial distributions around characteristic facility structures, indicating that MDCM encourages the network to learn more stable structural representations. This observation is consistent with the design motivation of the structural consistency score, which aims to preserve correlations among channels encoding related local structural patterns. Rather than enforcing explicit semantic-part alignment, MDCM acts as an implicit regularizer that alleviates unstable feature responses caused by noisy pseudosupervision. Compared with the baseline, the model equipped with MDCM exhibits more concentrated and structurally coherent feature activations, while irrelevant background responses are substantially suppressed. This result suggests that MDCM improves the robustness of feature representations under noisy pseudosupervision and provides complementary regularization from both structural and spectral perspectives.

F. Parameter Sensitivity Analysis

To evaluate the robustness of the proposed PKM, we further conduct parameter sensitivity experiments on two key hyperparameters, namely the number of part clusters K and the part size ratio $r = \{w_p/W, h_p/H\}$, where W and H denote the width and height of the original pseudobounding box, respectively.

We first investigate the influence of the number of part clusters K while fixing the part size ratio to 1/4. As shown in Table VII, the detection performance initially improves as K increases, indicating that moderate part decomposition helps the model capture more discriminative local structures. However, when K becomes excessively large, the performance starts to saturate or slightly decline due to overfragmentation and noisy part localization. Empirically, $K = 3$ achieves

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TABLE VII

SENSITIVITY ANALYSIS OF THE NUMBER OF PART CLUSTERS K WITH FIXED PART SIZE RATIO $r = 1/4$

Number of Parts K	mAP$_{50}$ (%)	mAP$_{50:95}$ (%)
1	71.3	35.6
2	71.9	35.3
3	71.8	35.8
4	70.3	34.0
5	71.6	34.4

TABLE VIII

SENSITIVITY ANALYSIS OF PART SIZE RATIO r WITH FIXED NUMBER OF CLUSTERS $K = 3$

Part Size Ratio r	mAP$_{50}$ (%)	mAP$_{50:95}$ (%)
1/6	70.9	35.4
1/5	71.3	34.9
1/4	71.8	35.8
1/3	70.4	34.4
1/2	68.1	34.0

the best balance between semantic diversity and localization stability.

We further analyze the influence of the part size ratio while fixing the number of clusters to $K = 3$. The results in Table VIII demonstrate that excessively small part regions may lose structural completeness and lead to unstable clustering, whereas overly large regions introduce redundant background information. A moderate part size ratio achieves the best trade-off between local discriminability and contextual integrity. In our experiments, the ratio $r = 1/4$ yields the most stable and effective performance.

Overall, these experiments verify that the proposed PKM is relatively robust to parameter variations within a reasonable range. The optimal configuration is achieved with $K = 3$ and $r = 1/4$, which is, therefore, adopted in all subsequent experiments.

G. Generalization Analysis

To further evaluate the generalization capability of the proposed framework, additional experiments are conducted on composite target datasets constructed from two public remote sensing benchmarks (DIOR and DOTA) as well as an industrial facility dataset (Indus-GF) collected from Chinese Earth observation satellites. These subsets contain heterogeneous objects with diverse spatial layouts and structural compositions. All experiments are performed under the same 25% labeled setting.

As reported in Table IX, the proposed PMC achieves the best overall performance on both composite benchmarks. On the DIOR-composite dataset, PMC attains an average mAP$_{50}$ of 84.8% and mAP$_{50:95}$ of 52.5%, outperforming all compared semisupervised methods. In particular, PMC achieves the highest performance on the HA category with 72.3% mAP$_{50}$, demonstrating its capability to recognize complex industrial composite targets with limited supervision. Compared with the baseline method Soft Teacher, PMC further improves the stricter mAP$_{50:95}$ from 51.0% to 52.5%, indicating more accurate localization. On the DOTA-composite dataset, PMC achieves the highest average mAP$_{50}$ of 63.9% and mAP$_{50:95}$ of 32.3%, exceeding representative teacher–student-based approaches such as Soft Teacher and DCST. Notably, PMC obtains a significant improvement on the AR category, reaching 84.2% mAP$_{50}$, which is substantially higher than other semisupervised methods. These results demonstrate that PMC can effectively generalize to complex composite targets across different datasets and maintain robust detection performance under diverse remote sensing scenarios. It is worth noting that the proposed PMC is specifically designed for composite-object detection, where object semantics are characterized by multiple discriminative parts and stable spatial configurations. Therefore, we evaluate the proposed framework on composite subsets of the DIOR and DOTA benchmarks, which provide representative composite targets for assessing part-aware representation learning. These subsets better match the intended application scenario of PMC than the complete benchmarks, which also contain many noncomposite object categories that do not exhibit stable part-level structures.

TABLE IX

GENERALIZATION PERFORMANCE ON COMPOSITE SUBSETS CONSTRUCTED FROM PUBLIC BENCHMARKS

Method	DIOR-composite					DOTA-composite			
	AR	ESA	HA	mAP$_{50}$	mAP$_{50:95}$	HA	AR	mAP$_{50}$	mAP$_{50:95}$
Supervised	82.3	89.4	65.7	79.1	46.4	39.7	54.7	47.2	19.6
DSL [13]	**88.2**	92.9	71.5	84.2	49.3	43.3	79.8	61.6	30.1
Active Teacher [18]	86.2	92.7	70.3	83.1	48.3	42.4	76.9	59.7	29.6
DCST [61]	83.9	90.3	71.6	82.0	48.5	42.0	75.2	58.6	27.3
Soft Teacher [16]	84.8	**94.6**	71.1	83.5	51.0	41.7	78.4	60.1	29.4
SSOD-AT [38]	85.6	94.2	70.9	83.6	50.8	**43.7**	76.2	60.0	28.8
Unbiased Teacher [59]	79.1	93.8	68.0	80.3	47.7	41.5	77.1	59.3	29.5
Unbiased Teacher v2 [60]	85.0	93.2	70.5	82.9	49.1	43.2	79.2	61.2	30.7
Consistent-Teacher [15]	80.9	67.5	45.3	64.6	34.3	23.9	77.2	50.6	21.6
Semi-DETR [20]	17.7	16.3	19.1	17.7	12.9	10.5	20.6	15.6	9.0
PMC (Ours)	88.0	93.9	**72.3**	**84.8**	**52.5**	43.5	**84.2**	**63.9**	**32.3**

TABLE X

GENERALIZATION PERFORMANCE ON THE INDUS-GF DATASET UNDER THE 25% LABELED SETTING

Method	TPP mAP$_{50}$ (%)	STP mAP$_{50}$ (%)	Avg mAP$_{50}$ (%)	mAP$_{50:95}$ (%)
Supervised	47.7	51.4	49.6	20.6
DSL [13]	60.4	53.2	56.8	21.7
Active Teacher [18]	61.7	51.0	56.4	21.2
DCST [61]	60.3	**54.6**	57.5	22.9
Soft Teacher [16]	62.8	51.1	57.0	25.3
SSOD-AT [38]	61.8	50.4	56.1	22.3
Unbiased Teacher [59]	56.4	49.7	53.1	21.3
Unbiased Teacher v2 [60]	65.7	51.3	58.5	24.0
Consistent-Teacher [15]	43.5	47.7	45.6	16.8
Semi-DETR [20]	11.0	11.1	11.1	6.5
PMC (Ours)	**67.5**	52.8	**60.2**	**26.5**

Beyond public benchmarks, the generalization performance of the proposed framework is further evaluated on the *Indus-GF* dataset. As shown in Table X, PMC consistently outperforms both supervised and semisupervised baselines under the same 25% labeled setting, achieving the highest average mAP$_{50}$ of 60.2% and mAP$_{50:95}$ of 26.5%. In particular, PMC achieves the best performance for TPP detection, reaching 67.5% mAP$_{50}$, which surpasses the previous best-performing method, Unbiased Teacher v2, by 1.8%. Meanwhile, PMC maintains competitive performance on STP detection, achieving 52.8% mAP$_{50}$. These results indicate that the proposed framework learns robust representations that can generalize effectively to imagery acquired from different satellite sensors with distinct spatial and spectral characteristics.

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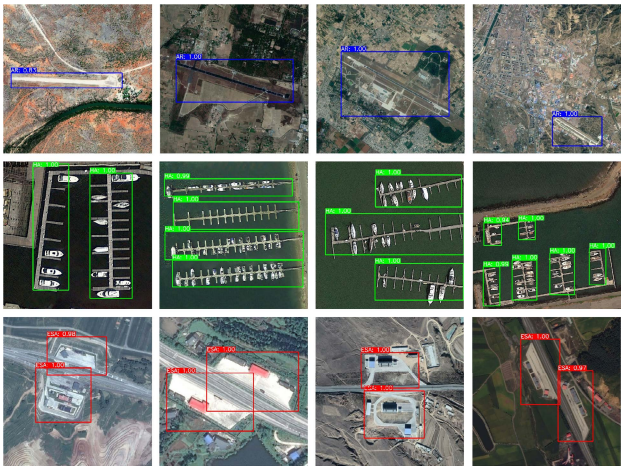


Fig. 7. Qualitative detection results of the proposed full model on the DIOR-composite dataset under the 25% labeled setting. (ESA, red bounding boxes; AR, blue bounding boxes; and HA, green bounding boxes).



Fig. 8. Qualitative detection results of the proposed full model on the DOTA-composite dataset under the 25% labeled setting. (AR, blue bounding boxes and HA, green bounding boxes).



Fig. 9. Qualitative detection results of the proposed full model on the Indus-GF dataset under the 25% labeled setting. (TPP, green bounding boxes, and STP, blue bounding boxes).

The consistent performance gains on DIOR-composite, DOTA-composite, and Indus-GF datasets indicate that the proposed framework is not tailored to a specific industrial scenario or data source, but instead learns transferable and structure-aware representations for composite target detection. This generalization capability is particularly important for practical large-scale applications, where annotated data are often scarce, and target characteristics vary across regions and sensor types.

In addition to quantitative evaluations, Figs. 7–9 present qualitative detection examples under the 25% labeled setting. The proposed method accurately localizes composite targets while preserving their internal structural composition, even in scenes with cluttered backgrounds, varying resolutions, and notable appearance discrepancies across datasets. These visual results further support the quantitative findings and demonstrate the robustness of the proposed approach across different benchmarks and satellite sources.

V. Conclusion

This article proposes PMC, a semisupervised part-aware detection framework with multidimensional consistency constraints for industrial composite object detection in RSIs. By introducing three complementary modules, the proposed method effectively addresses the challenges of insufficient structural representation, noisy pseudolabels, and the imbalance between pseudolabel quality and quantity in SSL. Extensive experiments demonstrate the effectiveness and generalization capability of PMC. Under a 25% labeled setting, PMC achieves 71.8% and 60.2% mAP on the Indus-CO and Indus-GF datasets, respectively, outperforming existing methods by 1.8%–14.1% and 2.7%–10.6%. These consistent improvements across different satellite platforms validate its robustness. Moreover, the consistent performance gains on composite subsets derived from the public DIOR and DOTA benchmarks further demonstrate the cross-dataset generalization capability of PMC for representative composite-object detection. Ablation studies verify the contribution of each component, where PKM, MDCM, and DTK improve the detection performance by 1.2%, 1.4%, and 0.6% mAP, respectively. Future work will focus on extending the proposed framework to a wider variety of composite-object categories and investigating its adaptation to more general remote sensing object detection scenarios, together with more advanced consistency learning strategies and cross-domain adaptation techniques.

References

[1] U. C. Mishra, "Environmental impact of coal industry and thermal power plants in India," *J. Environ. Radioactivity*, vol. 72, nos. 1–2, pp. 35–40, Jan. 2004.

[2] J. Liu, H. Chen, S. Zhao, P. Pan, L. Wu, and G. Xu, "Evaluation and improvements on the flexibility and economic performance of a thermal power plant while applying carbon capture, utilization & storage," *Energy Convers. Manage.*, vol. 290, Aug. 2023, Art. no. 117219.

[3] T. Liang et al., "Environmental impact evaluation of an iron and steel plant in China: Normalized data and direct/indirect contribution," *J. Cleaner Prod.*, vol. 264, Aug. 2020, Art. no. 121697.

[4] W. Tian et al., "CO_2 accounting model and carbon reduction analysis of iron and steel plants based on intra- and inter-process carbon metabolism," *J. Cleaner Prod.*, vol. 360, Aug. 2022, Art. no. 132190.

[5] L. Poudyal and K. Adhikari, "Environmental sustainability in cement industry: An integrated approach for green and economical cement production," *Resour., Environ. Sustainability*, vol. 4, Jun. 2021, Art. no. 100024.

[6] C. Guo, Z. Ma, Y. Zhao, C. Cao, Z. Jiang, and H. Zhang, "Multi-instance mining with dynamic localization for weakly supervised object detection in remote-sensing images," *Int. J. Remote Sens.*, vol. 46, no. 9, pp. 3487–3512, May 2025.

[7] H. Zhang, S. Wen, Z. Wei, and Z. Chen, "High-resolution feature generator for small-ship detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5617011.

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[8] N. Zhao, Y. Xian, T. Zhou, J. Shi, Z. Jiang, and H. Zhang, "Dense representative points-guided rotated-ship detection in remote sensing images," *Remote Sens.*, vol. 18, no. 3, p. 458, Feb. 2026.

[9] Q. He, B. Cheng, X. Zhang, and Y. Gan, "CFRANet: Cross-modal frequency-responsive attention network for thermal power plant detection in multispectral high-resolution remote sensing images," *Remote Sens.*, vol. 17, no. 15, p. 2706, Aug. 2025.

[10] P. F. Felzenszwalb, R. B. Girshick, D. McAllester, and D. Ramanan, "Object detection with discriminatively trained part-based models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 9, pp. 1627–1645, Sep. 2010.

[11] J. Zhu, X. Chen, and A. L. Yuille, "DeePM: A deep part-based model for object detection and semantic part localization," 2015, *arXiv:1511.07131*.

[12] X. Sun, P. Wang, C. Wang, Y. Liu, and K. Fu, "PBNet: Part-based convolutional neural network for complex composite object detection in remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 50–65, Mar. 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271620303555

[13] B. Chen, P. Li, X. Chen, B. Wang, L. Zhang, and X.-S. Hua, "Dense learning based semi-supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4805–4814.

[14] G. Luo, Y. Zhou, L. Jin, X. Sun, and R. Ji, "Towards end-to-end semi-supervised learning for one-stage object detection," 2023, *arXiv:2302.11299*.

[15] X. Wang et al., "Consistent-teacher: Towards reducing inconsistent pseudo-targets in semi-supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3240–3249.

[16] M. Xu et al., "End-to-end semi-supervised object detection with soft teacher," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3060–3069.

[17] G. Li, X. Li, Y. Wang, Y. Wu, D. Liang, and S. Zhang, "PseCo: Pseudo labeling and consistency training for semi-supervised object detection," in *Proc. 17th Eur. Conf. Comput. Vis. Cham*, Switzerland: Springer, 2022, pp. 457–472.

[18] P. Mi et al., "Active teacher for semi-supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 14482–14491.

[19] Y. Wang, L. Yao, G. Meng, X. Zhang, J. Song, and H. Zhang, "Addressing sample inconsistency for semisupervised object detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 6933–6944, 2024.

[20] J. Zhang et al., "Semi-DETR: Semi-supervised object detection with detection transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 23809–23818.

[21] P. Wang et al., "Omni-DETR: Omni-supervised object detection with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9357–9366.

[22] H. Zhang and Q. Deng, "Deep learning based fossil-fuel power plant monitoring in high resolution remote sensing images: A comparative study," *Remote Sens.*, vol. 11, no. 9, p. 1117, May 2019.

[23] Y. Zhou, F. Zhao, S. Wang, W. Liu, and L. Wang, "A method for monitoring iron and steel factory economic activity based on satellites," *Sustainability*, vol. 10, no. 6, p. 1935, Jun. 2018.

[24] Y. Yang, Y. Liu, L. Liu, Z. Liu, and H. Wu, "Monitoring global cement plants from space," *Remote Sens. Environ.*, vol. 302, Mar. 2024, Art. no. 113954.

[25] Z. Wang et al., "Detection of heavy-polluting enterprises from optical satellite remote sensing images," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2023, pp. 6474–6477.

[26] C. Ma, J. Yang, W. Xia, J. Liu, Y. Zhang, and X. Sui, "A model for expressing industrial information based on object-oriented industrial heat sources detected using multi-source thermal anomaly data in China," *Remote Sens.*, vol. 14, no. 4, p. 835, Feb. 2022.

[27] Y. Xie et al., "The potential of using SDGSAT-1 TIS data to identify industrial heat sources in the Beijing–Tianjin–Hebei region," *Remote Sens.*, vol. 16, no. 5, p. 768, Feb. 2024.

[28] Y. Zeng, R. Liao, C. Ma, D. Wang, and Y. Lv, "An approach to multiclass industrial heat source detection using optical remote sensing images," *Energies*, vol. 18, no. 4, p. 865, Feb. 2025.

[29] Y. Chen, M. Mancini, X. Zhu, and Z. Akata, "Semi-supervised and unsupervised deep visual learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 3, pp. 1327–1347, Mar. 2024.

[30] O. Chapelle, B. Scholkopf, and A. Zien, "Semi-supervised learning (chapelle, O. Et al., Eds.; 2006) [Book reviews]," *IEEE Trans. Neural Netw.*, vol. 20, no. 3, p. 542, Mar. 2009.

[31] X. J. Zhu, "Semi-supervised learning literature survey," *Dept. Comput. Sci., Univ. Wisconsin–Madison, Madison, WI, USA, Tech. Rep. 1530*, 2005.

[32] L.-Z. Guo, Z.-Y. Zhang, Y. Jiang, Y.-F. Li, and Z.-H. Zhou, "Safe deep semi-supervised learning for unseen-class unlabeled data," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 3897–3906.

[33] T. Shehzadi, D. Stricker, and M. Z. Afzal, "Semi-supervised object detection: A survey on progress from CNN to transformer," 2024, *arXiv:2407.08460*.

[34] Y. Chen, Y. Liu, Z. Gao, and J. Yan, "A novel semi-supervised learning framework for large-scale imbalanced remote sensing object detection," *Eng. Appl. Artif. Intell.*, vol. 163, Jan. 2026, Art. no. 113116.

[35] S. He, H. Zou, Y. Wang, X. Cao, H. Chen, and N. Jing, "Exploring cross-branch information for semi-supervised remote sensing object detection," *Eng. Appl. Artif. Intell.*, vol. 162, Dec. 2025, Art. no. 112378.

[36] W. Zha, L. Hu, C. Duan, and Y. Li, "Semi-supervised learning-based satellite remote sensing object detection method for power transmission towers," *Energy Rep.*, vol. 9, pp. 15–27, Sep. 2023.

[37] N. Liu, X. Xu, Y. Gao, Y. Zhao, and H.-C. Li, "Semi-supervised object detection with uncurated unlabeled data for remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 129, May 2024, Art. no. 103814.

[38] B. Zhang, Z. Wang, and B. Du, "Boosting semi-supervised object detection in remote sensing images with active teaching," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[39] J. Li, Y. Ji, H. Xu, K. Cheng, R. Song, and Q. Du, "UAT: Exploring latent uncertainty for semi-supervised object detection in remote-sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5633212.

[40] X. Zhu et al., "Temporal-feedback self-training for semi-supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5621614.

[41] Z. Fang, J. Ren, J. Zheng, R. Chen, and H. Zhao, "Dual teacher: Improving the reliability of pseudo labels for semi-supervised oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5602515.

[42] Z. Lin, G. Hua, and L. S. Davis, "Multiple instance feature for robust part-based object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 405–412.

[43] B. Wu and R. Nevatia, "Improving part based object detection by unsupervised, online boosting," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2007, pp. 1–8.

[44] Y. Tang, X. Wang, E. Dellandrea, and L. Chen, "Weakly supervised learning of deformable part-based models for object detection via region proposals," *IEEE Trans. Multimedia*, vol. 19, no. 2, pp. 393–407, Feb. 2017.

[45] M. Sun and S. Savarese, "Articulated part-based model for joint object detection and pose estimation," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 723–730.

[46] W. Zhang, X. Sun, K. Fu, C. Wang, and H. Wang, "Object detection in high-resolution remote sensing images using rotation invariant parts based model," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 74–78, Jan. 2014.

[47] J. Li, H.-C. Wong, S.-L. Lo, and Y. Xin, "Multiple object detection by a deformable part-based model and an R-CNN," *IEEE Signal Process. Lett.*, vol. 25, no. 2, pp. 288–292, Feb. 2018.

[48] T. Fang, D. Fang, and B. Yu, "Carbon emission efficiency of thermal power generation in China: Empirical evidence from the micro-perspective of power plants," *Energy Policy*, vol. 165, Jun. 2022, Art. no. 112955.

[49] K. Li, H. Fan, and P. Yao, "Estimating carbon emissions from thermal power plants based on thermal characteristics," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 128, Apr. 2024, Art. no. 103768.

[50] T. Lei et al., "Global iron and steel plant CO_2 emissions and carbon-neutrality pathways," *Nature*, vol. 622, no. 7983, pp. 514–520, Oct. 2023.

[51] J. Zhang, J. Shen, L. Xu, and Q. Zhang, "The CO_2 emission reduction path towards carbon neutrality in the Chinese steel industry: A review," *Environ. Impact Assessment Rev.*, vol. 99, Mar. 2023, Art. no. 107017.

[52] Y. Izumi, A. Iizuka, and H.-J. Ho, "Calculation of greenhouse gas emissions for a carbon recycling system using mineral carbon capture and utilization technology in the cement industry," *J. Cleaner Prod.*, vol. 312, Aug. 2021, Art. no. 127618.

[53] C. Tan, X. Yu, and Y. Guan, "A technology-driven pathway to net-zero carbon emissions for China's cement industry," *Appl. Energy*, vol. 325, Nov. 2022, Art. no. 119804.

[54] F. Han et al., "Improvement of monitoring production status of iron and steel factories based on thermal infrared remote sensing," *Sustainability*, vol. 15, no. 11, p. 8575, May 2023.

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[55] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271619302825

[56] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[57] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 1–9.

[58] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[59] Y.-C. Liu et al., "Unbiased teacher for semi-supervised object detection," 2021, *arXiv:2102.09480*.

[60] Y.-C. Liu, C.-Y. Ma, and Z. Kira, "Unbiased teacher V2: Semi-supervised object detection for anchor-free and anchor-based detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9809–9818.

[61] K. Wang et al., "Double-check soft teacher for semi-supervised object detection," in *Proc. 31st Int. Joint Conf. Artif. Intell.*, Jul. 2022, pp. 1430–1436.



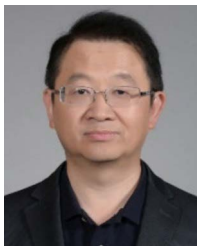
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